# Frontend: Theory Assignment 2

**Name : K.Bala Sai Praveen**

**Univ roll: 22515000067**

**NB-20**

1. **What is CSS Box Model?**

The CSS Box Model explains how every HTML element is treated as a rectangular box on a webpage.
It consists of four parts:

* Content – The actual text or element.
* Padding – Space between the content and border.
* Border – A line around the padding and content.
* Margin – Space outside the border.

Example:

```
div {
        width: 200px;
        padding: 20px;
        border: 2px solid black;
        margin: 10px;
}
```

the Box Model helps us control the **size and spacing of elements** on a webpage.

2. **What is Media Query? Explain with sample code.**

A media query is a CSS feature used to make a webpage responsive.
It allows us to apply different CSS styles depending on the screen size, device, or other conditions.

For example, we can change the background color when the screen becomes smaller.

```
body {
background-color: white;
}

@media (max-width: 600px) {
  body {
    background-color: lightblue;
  }
}
```

Here, when the screen width is **600px or less**, the background becomes light blue.

Media queries help websites look good on **mobiles, tablets, and computers**.

3. **What are the features of Bootstrap?**

**Bootstrap** is a popular frontend framework used to create responsive and attractive websites quickly.
Some important features are:

**Responsive design** – Websites automatically adjust to different screen sizes.

1. **Grid system** – Helps arrange content into rows and columns.
2. **Predefined components** – Provides buttons, cards, navigation bars, forms, etc.
3. **Easy to use** – We can create designs using ready-made classes.
4. **Mobile-first approach** – It is designed with smaller screens in mind.
5. **Cross-browser compatibility** – Works across different modern browsers.

4. **Write an example of basic grid structure in Bootstrap**

A basic grid structure is:

```
      <div class="container">
<div class="row">
  <div class="col-6">
    Column 1
  </div>
```

```
    <div class="col-6">
      Column 2
    </div>
  </div>
</div>
```

Here:

- container creates the main area.
- row creates a row.
- col-6 takes **6 out of 12 columns**.
- Two col-6 elements together occupy the complete row.

5. **What are CSS Selectors?**

**CSS selectors** are used to select HTML elements so that we can apply styles to them.

For example:

```
p {
  color: blue;
}
```

Here, p is a selector, so the style is applied to all <p> elements.

| Selector | Example | Meaning |
|---|---|---|
| Element | p | Selects all <p> elements |
| Class | .box | Selects elements with class box |
| ID | #header | Selects an element with ID header |
| Universal | * | Selects all elements |

6. **What are CSS Backgrounds? List the properties.**

CSS backgrounds are used to add colors or images behind HTML elements.

Some important background properties are:

- background-color
- background-image
- background-repeat

- background-position
- background-size
- background-attachment

```
body {
  background-color: lightblue;
  background-image: url("image.jpg");
  background-size: cover;
}
```

**7. How can we overlap elements in CSS?**

We can overlap elements by using positioning properties and controlling their position with top, left, right, or bottom.
The z-index property can also be used to decide which element appears on top.
Example:

```
.box1 {
  position: absolute;
  top: 50px;
  left: 50px;
}

.box2 {
  position: absolute;
  top: 80px;
  left: 80px;
  z-index: 2;
}
```

**8. What are the different positioning properties in CSS?**

CSS provides different ways to position elements.

The main positioning values are:

1. **Static** – Default position of an element.
2. **Relative** – Moves an element relative to its normal position.
3. **Absolute** – Positions an element relative to its nearest positioned parent.
4. **Fixed** – Keeps an element fixed relative to the browser window.
5. **Sticky** – Acts like a normal element until a certain scroll position, then behaves like a fixed element.

**Example:**

```
        .box{
  position: relative;
  top: 20px;
  left: 10px;
}
```

**9. What is ECMAScript?**

**ECMAScript** is a standard that defines the rules and features of the **JavaScript programming language**.

JavaScript follows the ECMAScript standard.

It defines things such as:

- Variables
- Functions
- Data types
- Loops
- Classes
- Modern JavaScript features

**In simple words:** ECMAScript is the **standard or specification behind JavaScript**.

**10. Difference between var, let and const in JavaScript. Explain Temporal Dead Zone.**

| Feature | var | let | const |
|---|---|---|---|
| Scope | Function scope | Block scope | Block scope |
| Re-declaration | Allowed | Not allowed in same scope | Not allowed in same scope |
| Re-assignment | Allowed | Allowed | Not allowed |

| Feature | var | let | const |
| --- | --- | --- | --- |
| Hoisting | Hoisted and initialized as undefined | Hoisted but not initialized | Hoisted but not initialized |